DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Design) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Computer Science and Design) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCSD501 - THEORY OF COMPUTATION

Time : 3 Hours] **[Total Marks : 60**

Instructions to the Students:

1.Each question carries 12 marks.

2.Question No. 1 will be compulsory and include objective-type questions.

3.Candidates are required to attempt any four questions from Question No. 2 to Question No.6

4.Use of non-programmable scientific calculators is allowed.

5.Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question) 12

1 Which of the following is true?
a) (01)*0 = 0(10)* b) (0+1)*0(0+1)*1(0+1) = (0+1)*01(0+1)*
c) (0+1)*01(0+1)*+1*0* = (0+1)* d) All of the mentioned

2 Regular expression for all strings starts with ab and ends with bba is.
a)aba*b*bba b) ab(a+b)*bba c) ab(ab)*bba d)All of the mentioned

3 Reverse of (0+1)* will be
a) Null b) (0+1) c) (0+1)* d)0+1)(0+1)*

4 Which of the following is an application of Finite Automaton?
a)Compiler Design b) Grammar Parsers c) Text Search d) All of the mentioned

5 5.Which of the following statement is false?
a) Context free language is the subset of context sensitive language
b) Regular language is the subset of context sensitive language
c) Recursively ennumerable language is the super set of regular language
d) Context sensitive language is a subset of context free language

6 6.The format: A->aB refers to which of the following?
a) Chomsky Normal Form b) Greibach Normal Form
c) Backus Naur Form d) None of the mentioned

7 Consider G=({S,A,B,E}, {a,b,c},P,S), where P consists of S →AB, A →a, B →b and E →c. Number of productions
in P' after removal of useless symbols:
a) 4 b) 3 c) 2 d) 5

8 The language accepted by Push down Automaton is
a) Recursive b) Context free c) Linearly Bounded d) All of the mentioned

9 The language accepted by Finite Automaton is
a) Regular b) Context free c) Linearly Bounded d) All of the mentioned

10 A push down automaton employs ________ data structure.
a) Queue b) Linked List c) Hash Table d) Stack

11 For S->0S1|ϵ for ∑={0,1}*, which of the following is wrong for the language produced?
a) Non regular language b) 0^n1^n| n>=0 c) 0^n1^n | n>=1 d) None of the mentioned

12 Are ambiguous grammar context free?
a) Yes b) No

Q2. Solve the following.

A) Design Deterministic Finite Automaton that accepts even number of a's over alphabet "a". 6

B) Find the shortest string in the language, represented by the regular expression a*(ab)*b*. 6

Q3. Solve the following.

A) Design Non Deterministic Finite Automatonthat accepts set of all strings over {0,1} that have atleast two consecutive 0's or 1's. 6

B) State some applications of regular expressions and finite automata. 6

Q4. Solve Any Two of the following.

A) Explain the steps to convert Mealy machine to Moore machine . Distinguish between Moore machine and Mealy machine. 6

B) Define the following and write example of each. 6
(i) Regular grammar (ii)Recursive grammar (iii)Linear and non linear grammar

C) Consider the following grammar . 6
S ---> aB|bA
A---> a|aS|bAA
B --> b|bS| aBB
Obtain leftmost derivation and righmost derivation and parse trees for leftmost derivation and righmost derivation for string *aaabbabbba.*

Q5. Solve Any Two of the following.

A) Distinguish between deterministic and non deterministic Pushdown Automaton. 6

B) Find the reduced grammar equivalent to the following grammar. 6
S--> aB | bX
A--> BAd | bSX | q
B--> aSB | bBX
X--> SBD | aBX | ad

C) Prove that the following grammar is ambiguous grammar. 6
E--> a | b
E--> E -E

Q6. Solve Any Two of the following.

A) Explain the components of Pushdown automaton. 6

B) Distinguish between tape and head of Turing machine and Pushdown Automaton. 6

C) Design Turing machine to accept the language of 0's and 1's ending with 00. 6

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